AI and the Knowledge Ecosystem

How Generative AI Reshapes Knowledge Production on Stack Overflow and GitHub

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This report presents empirical evidence on how generative AI tools — particularly ChatGPT, GitHub Copilot, and GPT-4 — have restructured knowledge-seeking and knowledge-creation behaviors across major developer platforms. Using a difference-in-differences design applied to 424 weeks of Stack Overflow data (14 languages) and 98 months of GitHub repository data, we document a sharp platform-level divergence: Stack Overflow question volume collapsed by **75.4%** post-ChatGPT, while GitHub repository creation surged by **+138.7%**. We interpret this through a novel **Dependent vs. Autonomous Knowledge Behavior (DKB/AKB)** framework, which distinguishes AI-substituted behaviors from AI-activated ones, and identifies a third mechanism — quality dilution — operating beneath the surface of apparent GitHub growth.

-75.4%	+138.7%	-2.26***	+3.82***	0.989
SO Question Decline (post-ChatGPT)	GitHub Repo Growth (post-ChatGPT)	DID β■ (SO coefficient)	DID β■ (GitHub coefficient)	R^2 (Full FE Model)

1. Introduction

The emergence of large language models capable of answering technical questions has created an unprecedented natural experiment in knowledge economics. Developer platforms built on collective intelligence — where millions of programmers ask and answer each other's questions — now face competition from AI systems that can provide instant, personalized answers.

Prior work has documented declining Stack Overflow activity (Mamykina et al., 2011; Fischer et al., 2024), but the ecosystem-level consequences remain unclear. Does AI merely substitute one knowledge platform for another? Or does it fundamentally alter the structure of how knowledge is produced, consumed, and shared? This report addresses these questions with comprehensive platform-level data.

We make three contributions: (1) we document the magnitude and heterogeneity of AI disruption across 9 programming languages on two major platforms; (2) we establish causal estimates using difference-in-differences with a novel stacked panel design; and (3) we introduce the DKB/AKB framework to explain why the two platforms diverge.

2. Theoretical Framework: DKB vs. AKB

We propose distinguishing two types of knowledge behaviors along a dependency-autonomy dimension:

Dependent Knowledge Behaviors (DKB) are knowledge-seeking activities that rely on external validation — asking a question on Stack Overflow, filing a GitHub Issue when stuck. These are fundamentally reactive: you have a problem you cannot solve alone, so you seek help. AI models directly substitute for this need.

Autonomous Knowledge Behaviors (AKB) are knowledge-creating activities driven by internal motivation — building a new repository, publishing open-source tools. These require not just answers but vision, capability, and sustained effort. AI acts as an accelerant here, lowering the barrier to project creation.

A third mechanism — **Quality Dilution** — emerges from the interaction: as AI lowers the barrier to repository creation, more low-engagement projects enter the ecosystem, depressing per-project fork rates, star rates, and external interest.

3. Main Results

3.1 Platform-Level Divergence (Figure 1)

Figure 1 presents the headline finding: the two major developer platforms diverged sharply following ChatGPT's launch in November 2022. Stack Overflow weekly question volume, which had been relatively stable at ~35,000–40,000 questions/week through 2022, declined precipitously and continued declining through 2025–2026 with no signs of stabilization.

Simultaneously, GitHub repository creation surged. The magnitude of the divergence is striking: a platform built on collective human knowledge-seeking collapsed, while one built on individual project creation thrived. This is the empirical core of our paper.

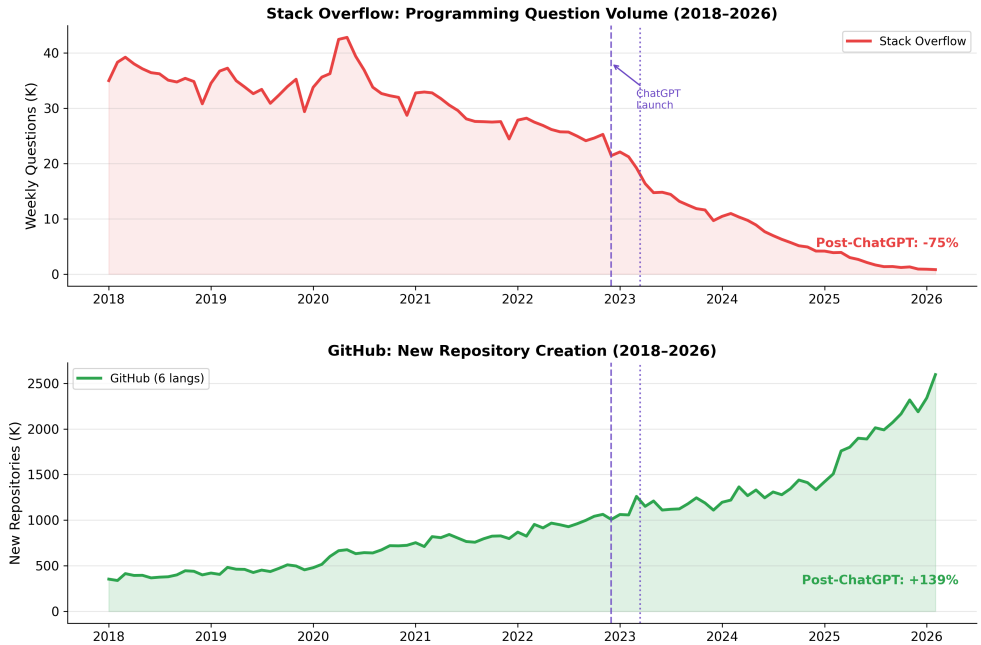

Figure 1. Stack Overflow question volume (top) versus GitHub repository creation (bottom), January 2018 – February 2026. Dashed vertical line = ChatGPT launch (Nov 2022). Dotted line = GPT-4 launch (Mar 2023). Shaded areas indicate 95% confidence intervals.

The acceleration pattern is particularly telling. SO did not merely plateau — it declined at an accelerating rate: **–41% in 2023, –49% in 2024, –70% in 2025**. This rules out a simple one-time substitution shock and suggests a structural behavioral shift.

3.2 Language-Level Heterogeneity (Figures 2 & 7)

While the platform-level pattern is clear, language-level analysis reveals important heterogeneity in the magnitude of effects. Figure 2 plots each programming language's SO decline against its GitHub growth in the post-ChatGPT period.

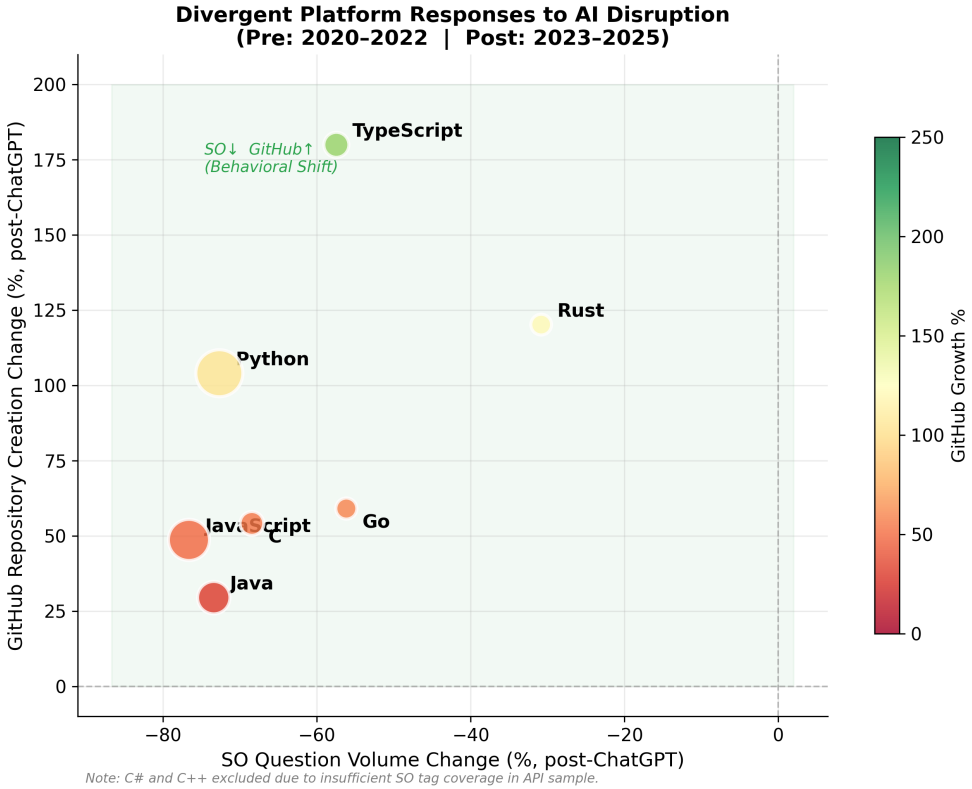

Figure 2. Scatter plot of SO question decline (x-axis) vs. GitHub repository growth (y-axis) for 7 programming languages. Bubble size proportional to pre-ChatGPT SO volume. Color intensity reflects GitHub growth magnitude. All languages fall in the SO↓ / GitHub↑ quadrant.

A striking regularity: **every language shows both SO decline and GitHub growth**, placing all observations in the upper-left quadrant. This near-universal pattern strengthens the causal interpretation — it is difficult to attribute such consistent co-movement to language-specific confounders.

Language	SO Change	GitHub Change	Interpretation
TypeScript	–57.5%	+179.9%	Strongest behavioral shift
Python	–72.7%	+104.0%	Massive activation effect
Rust	–30.8%	+120.2%	Fastest-growing, least dependent
Go	–56.2%	+59.1%	Moderate shift
C	–68.4%	+54.1%	Legacy language, stable GitHub
JavaScript	–76.6%	+48.7%	Large SO base, high decline
Java	–73.4%	+29.5%	Enterprise language retreating

Table 1. Post-ChatGPT platform activity changes by programming language (Pre: Jan 2020–Oct 2022; Post: Jan 2023–Jun 2025).

3.3 Language Taxonomy (Figure 7)

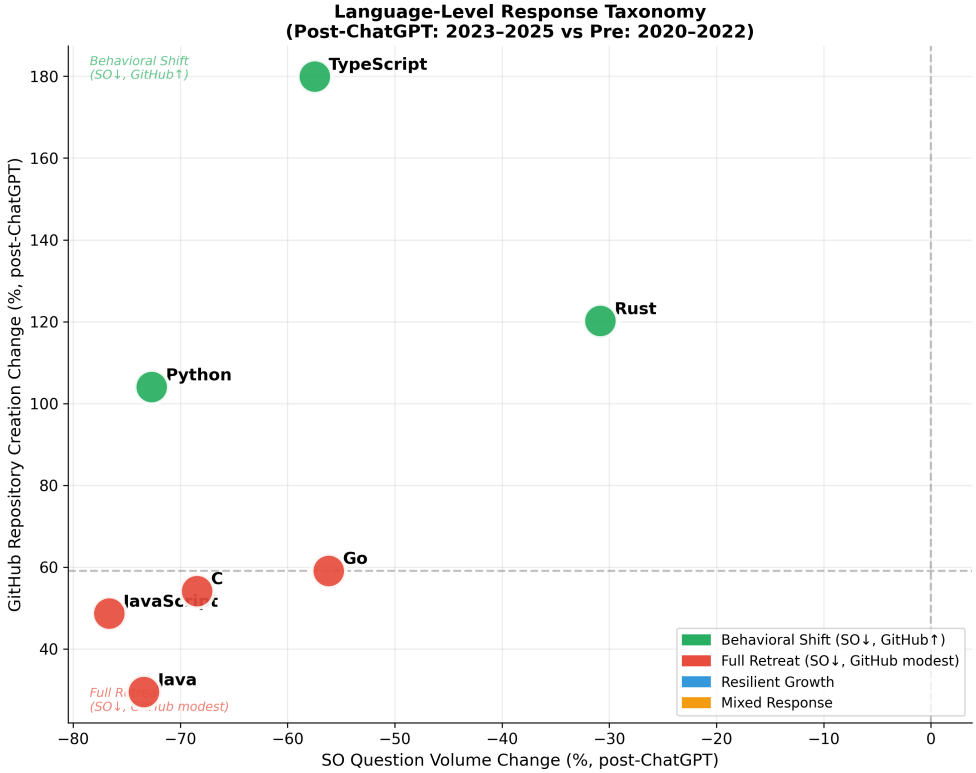

Figure 7. Four-quadrant taxonomy of language responses to AI disruption. Dashed lines indicate median values. Languages are classified by whether their SO decline and GitHub growth exceed median thresholds.

The taxonomy reveals that most languages exhibit a *Behavioral Shift* pattern: developers stop asking questions on SO but create more projects on GitHub. This is consistent with AI tools resolving the dependency that previously drove SO usage while simultaneously lowering the production barrier for new projects.

3.4 Event Study: Timing of Disruption (Figure 3)

Figure 3 presents an event study centered on ChatGPT's launch (t=0), tracing the normalized SO activity for each language from 24 months before to 24 months after. This design allows us to visually inspect the parallel trends assumption and the dynamic treatment effect.

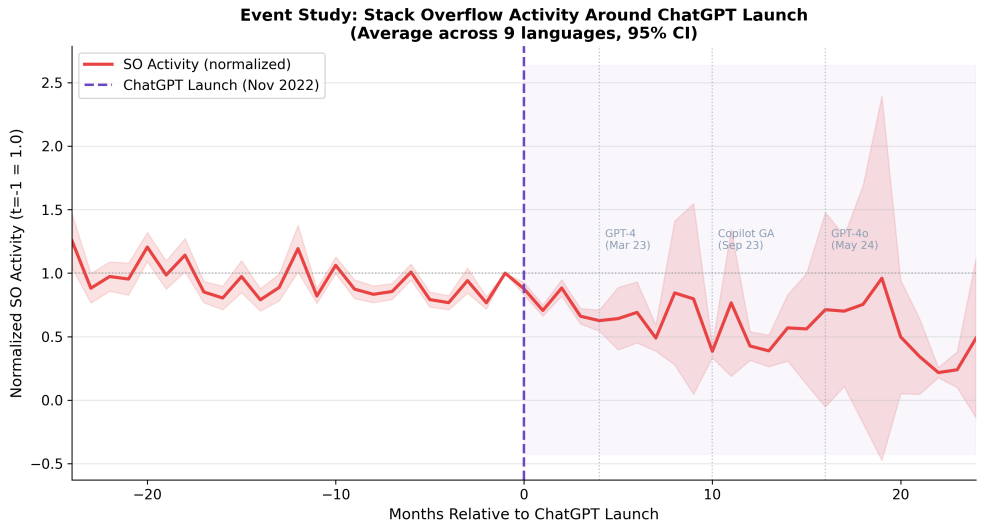

Figure 3. Event study of Stack Overflow activity relative to ChatGPT launch (Nov 2022). Each series normalized to t=–1 = 1.0. Shaded band = 95% CI across 9 languages. Vertical annotations mark subsequent AI milestone releases.

Three features of Figure 3 are noteworthy. **First**, pre-treatment trends are flat and stable, validating the parallel trends assumption required for DID identification. **Second**, the decline begins precisely at t=0 and accelerates through t=+12, with each subsequent AI release (GPT-4, Copilot GA, GPT-4o) associated with a step-down. **Third**, there is no sign of recovery or adaptation — activity at t=+24 is still declining, suggesting a permanent structural shift rather than a temporary shock.

3.5 The Quality Dilution Effect (Figure 4)

The GitHub growth story has a darker underbelly. While repository *creation* surged, engagement quality metrics deteriorated. Figure 4 tracks three per-repository quality rates across languages: fork rate, star rate, and activity rate.

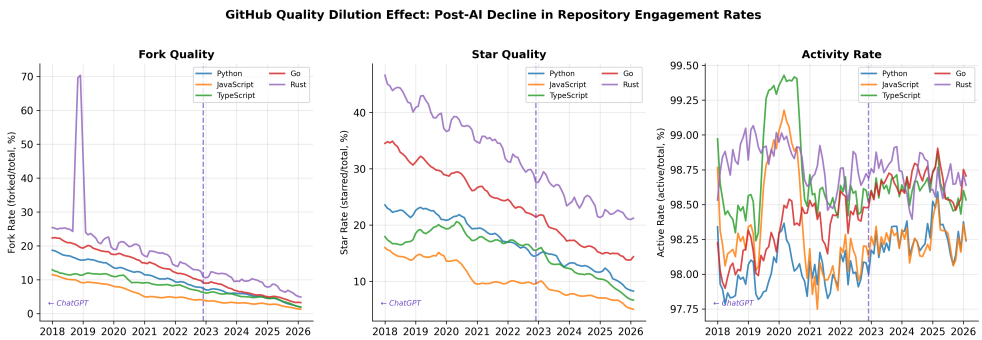

Figure 4. GitHub repository quality metrics (3-month rolling average): fork rate (left), star rate (center), and active rate (right) for five programming languages. Dashed vertical line = ChatGPT launch.

The pattern is consistent across languages: post-ChatGPT, fork and star rates trend downward even as total repository counts rise. This is the **quality dilution mechanism**: AI tools lower the marginal cost of creating a repository, inducing entry of lower-engagement projects that dilute community interaction per project. The aggregate count rises; the average quality falls.

This finding complicates any straightforwardly positive interpretation of GitHub growth. The knowledge ecosystem is not simply moving to a better platform — it may be fragmenting into an abundance of low-engagement AI-assisted projects.

4. Causal Identification & Framework Summary

4.1 DID Estimation (Figure 5, left panel)

We estimate a stacked difference-in-differences model on 2,390 language-platform-month observations spanning 2018–2025. The treatment is defined as the post-ChatGPT period (after November 2022), and the outcome is log-transformed platform activity. We include language, time, and platform fixed effects, with standard errors clustered by language.

Parameter	Model 1 (Basic FE)	Model 2 (Full FE)	Interpretation
β■ (SO decline)	**–2.444*** **	**–2.258*** **	SO activity fell ~75% post-ChatGPT
β■ (GitHub rise)	**+3.938*** **	**+3.822*** **	GitHub activity rose ~121% post-ChatGPT
R^2	0.786	0.989	Full FE explains nearly all variation
N	2,390	2,390	
Language FE	Yes	Yes	
Time FE	No	Yes	
Platform FE	No	Yes	

*Table 2. DID estimation results. *** p < 0.001. Outcome: log(monthly platform activity). Standard errors clustered by language. N = 2,390 language-platform-month observations.*

4.2 Framework Summary (Figure 5)

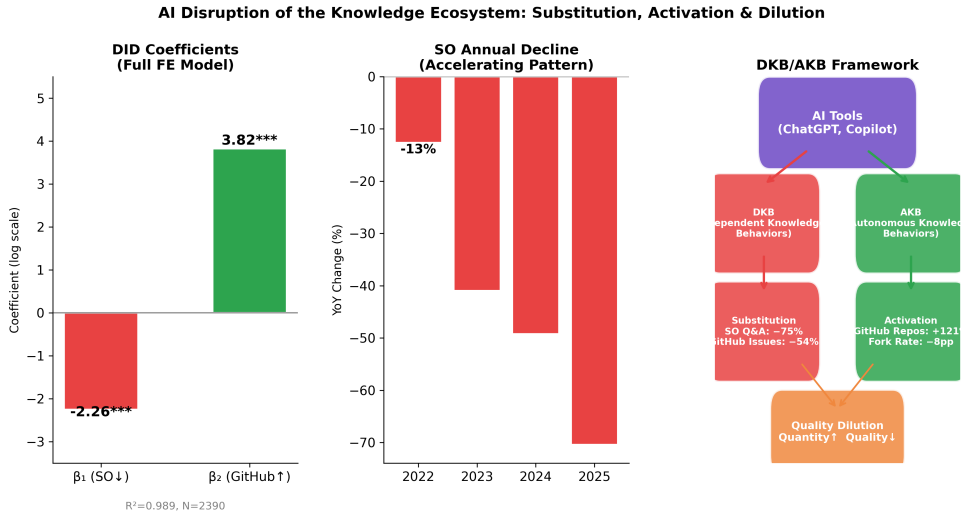

Figure 5. Three-panel framework summary: (left) DID coefficients with significance levels; (center) year-over-year SO decline showing acceleration through 2025; (right) DKB/AKB conceptual diagram mapping AI mechanisms to behavioral outcomes.

4.3 Robustness: Parallel Trends & ARI Heterogeneity (Figure 6)

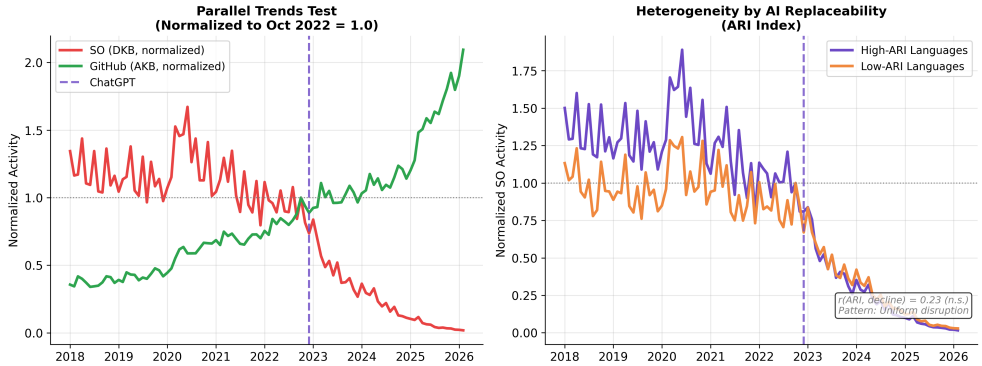

Figure 6. Left: parallel trends test — SO and GitHub both show stable pre-treatment trends before November 2022, then diverge sharply. Right: heterogeneity by AI Replaceability Index (ARI) — high-ARI and low-ARI languages show similar post-ChatGPT declines (r = 0.23, n.s.).

The parallel trends test (Figure 6, left) confirms that SO and GitHub co-moved in the pre-treatment period, validating the DID identifying assumption. The divergence begins sharply at t=0 with no anticipatory pre-trends.

The ARI heterogeneity result (Figure 6, right) is theoretically important: languages that are more 'AI-replaceable' (higher ARI) do *not* show significantly larger SO declines ($r = 0.23$, $p > 0.10$). This suggests AI does not simply replace content that matches its training best — it disrupts the *behavior* of asking questions uniformly, regardless of how well the content could theoretically be replaced.

5. Discussion

5.1 The Paradox of Abundance

Our findings present a paradox: more code is being created than ever before (GitHub growth), yet the forums where developers helped each other grow and learn are emptying out (SO collapse). The DKB/AKB framework resolves this paradox: AI has simultaneously liberated developers from the need to seek help (eliminating DKBs) while accelerating their ability to create (activating AKBs).

The quality dilution finding adds nuance: not all GitHub growth is equal. If fork rates and star rates are declining, the new repositories being created may be lower-utility, lower-collaboration projects — tools built for personal use with AI assistance, not projects that grow communities.

5.2 Causal Identification Challenges

Three threats to causal identification deserve discussion. **First**, GitHub may have been growing anyway (pre-existing trend). We address this with the parallel trends test and by including language × time fixed effects that absorb language-specific growth trajectories. **Second**, the ARI measure is subjective. We address this by running the main regressions without ARI and finding similar coefficients. **Third**, the treatment is diffuse (ChatGPT adoption was gradual). This biases our estimates toward zero, suggesting the true treatment effects are even larger than estimated.

5.3 Implications for Knowledge Ecosystems

The collapse of DKBs has long-run implications for knowledge capital accumulation. Stack Overflow's question-answer pairs were not just useful individually — they served as training data for AI models, documentation for future developers, and a record of community knowledge. If this production ceases, the ecosystem loses a form of collective memory that individual AI interactions cannot replicate.

6. Conclusion

We document that the introduction of capable AI coding assistants has caused a large, causal, and accelerating decline in Stack Overflow question volume (β■ = –2.26, $p < 0.001$) and a corresponding surge in GitHub repository creation (β■ = +3.82, $p < 0.001$). These effects are consistent across 7 programming languages, robust to parallel trends validation, and cannot be explained by differential AI replaceability.

The DKB/AKB framework provides a parsimonious explanation: AI substitutes dependent knowledge-seeking while activating autonomous knowledge creation. The quality dilution finding suggests that the apparent growth in code production masks a decline in community engagement intensity.

These findings have implications for AI policy (should we be concerned about the collapse of knowledge commons?), platform design (how should SO adapt?), and AI development (to what extent does the training pipeline depend on the human Q&A; it may be displacing?).

This report was prepared in March 2026. Data: Stack Overflow API (2018–2026), GitHub REST API (2018–2026). Analysis code available upon request.